

ARTIFICIAL INTELLIGENCE

Lecture Set 11
Hidden Markov Model

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Course Contents

1. Introduction to AI
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- **11. Hidden Markov model**
12. Markov Decision Process
13. Machine Learning (Supervised, Unsupervised, Reinforcement Learning)

Probability Recap

- Conditional probability $P(x|y) = \frac{P(x, y)}{P(y)}$
- Product rule $P(x, y) = P(x|y)P(y)$
- Chain rule
$$\begin{aligned} P(X_1, X_2, \dots, X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots \\ &= \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$
- X, Y independent if and only if: $\forall x, y : P(x, y) = P(x)P(y)$
- X and Y are conditionally independent given Z if and only if:

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

$$X \perp\!\!\!\perp Y | Z$$

Reasoning over Time or Space

- Often, we want to reason about a *sequence* of observations where the state of the underlying system is *changing*
 - Speech recognition
 - Robot localization
 - User attention
 - Medical monitoring
 - Global climate
- Need to introduce time into our models

Markov assumption

- **Markov assumption:** The assumption that the current state depends on only a finite fixed number of previous states.
- **Markov chain:** a sequence of random variables where the distribution of each variable follows the Markov assumption

Markov assumption

One simplifying yet extremely powerful notion

Markov property

Ex: can you predict the next letter?

I

I a

I am

I am t

...

I am talking about Markov

At any given location in the text stream, the next letter to appear depends on only a few letters back, not the whole history

Markov assumption

State variable

Ex: identity of the letter $s = \{s^0, s^1, s^2, \dots, s^{26}\}$

At time t , the state variable is denoted by s_t

Markov property

$$p(s_t | s_1, s_2, \dots, s_{t-1}) = p(s_t | s_{t-1})$$

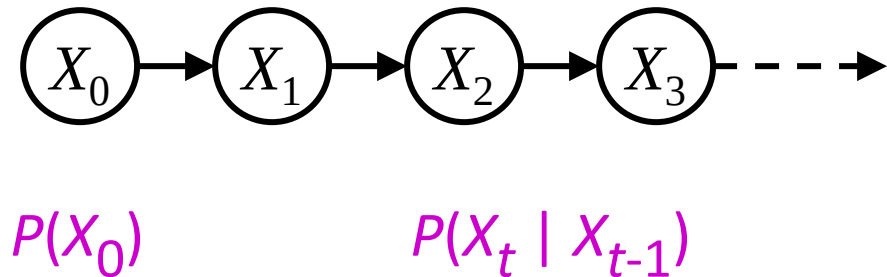
$p(s_t | s_{t-1})$

referred as **transition** probabilities

there are 27×27 of them for transitions between letters.

Markov Models (aka Markov chain/process)

- Value of X at a given time is called the **state** (usually discrete, finite)



- The **transition model** $P(X_t | X_{t-1})$ specifies how the state evolves over time
- Stationarity** assumption: transition probabilities are the same at all times
- Markov** assumption: “future is independent of the past given the present”
 - X_{t+1} is independent of $X_0, \dots, X_{t-1}$ given X_t
 - This is a **first-order** Markov model (a k th-order model allows dependencies on k earlier steps)
- Joint distribution $P(X_0, \dots, X_T) = P(X_0) \prod_t P(X_t | X_{t-1})$

Markov Models (aka Markov chain/process)

First-order Markov process: the current state depends only on the previous state and not on any earlier states.

$$P(X_t | X_{0:t-1}) = P(X_t | X_{t-1})$$

Current t-1 state provides enough information to make the future conditionally independent of the past,

Second-order Markov process: The transition model for a second-order Markov process is the conditional distribution

$$P(X_t | X_{t-2}, X_{t-1})$$

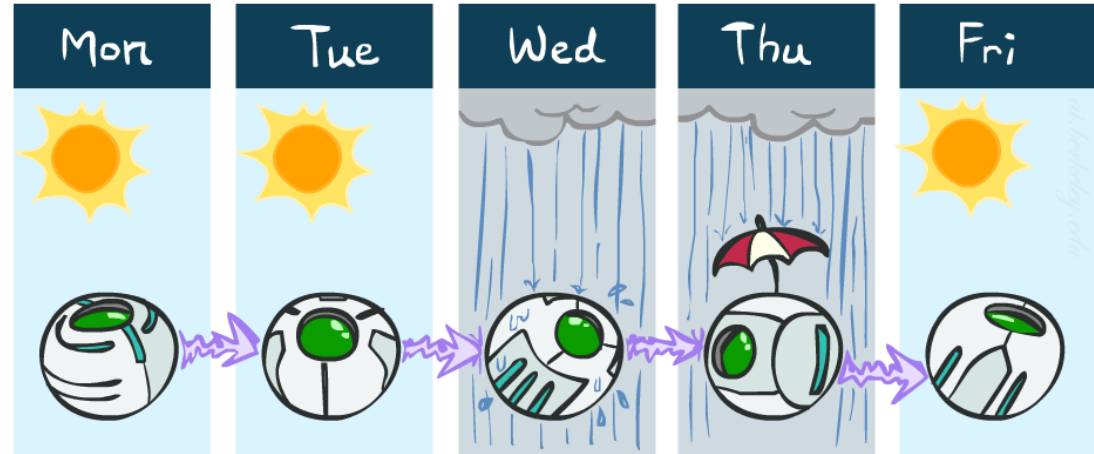
Sensor Markov assumption (observation model)

$$P(E_t | X_{0:t}, E_{0:t-1}) = P(E_t | X_t)$$

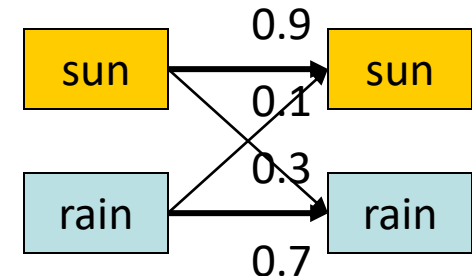
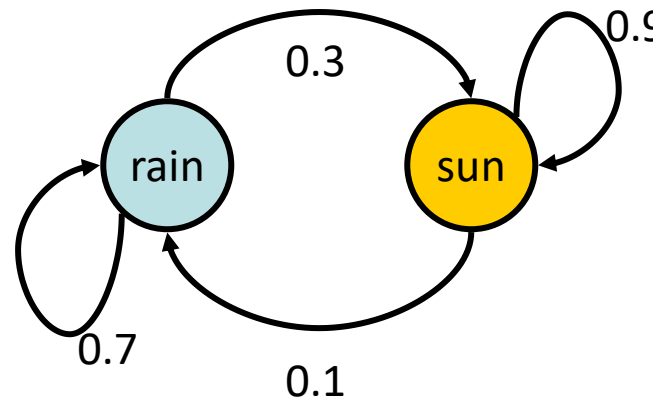
Example Markov Chain: Weather

- States: $X = \{\text{rain}, \text{sun}\}$
- Initial distribution: 1.0 sun
- CPT $P(X_t \mid X_{t-1})$:

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

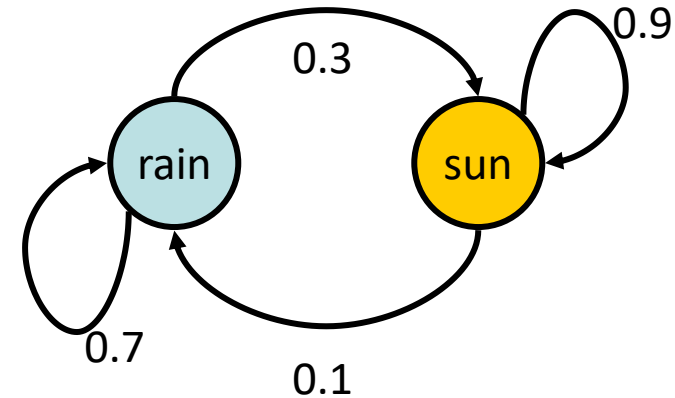


Two new ways of representing the same CPT



Example Markov Chain: Weather

- Initial distribution: 1.0 sun



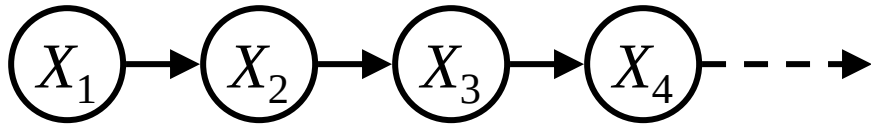
- What is the probability distribution after one step?

$$P(X_2 = \text{sun}) = P(X_2 = \text{sun} | X_1 = \text{sun})P(X_1 = \text{sun}) + P(X_2 = \text{sun} | X_1 = \text{rain})P(X_1 = \text{rain})$$

$$0.9 \cdot 1.0 + 0.3 \cdot 0.0 = 0.9$$

Mini-Forward Algorithm

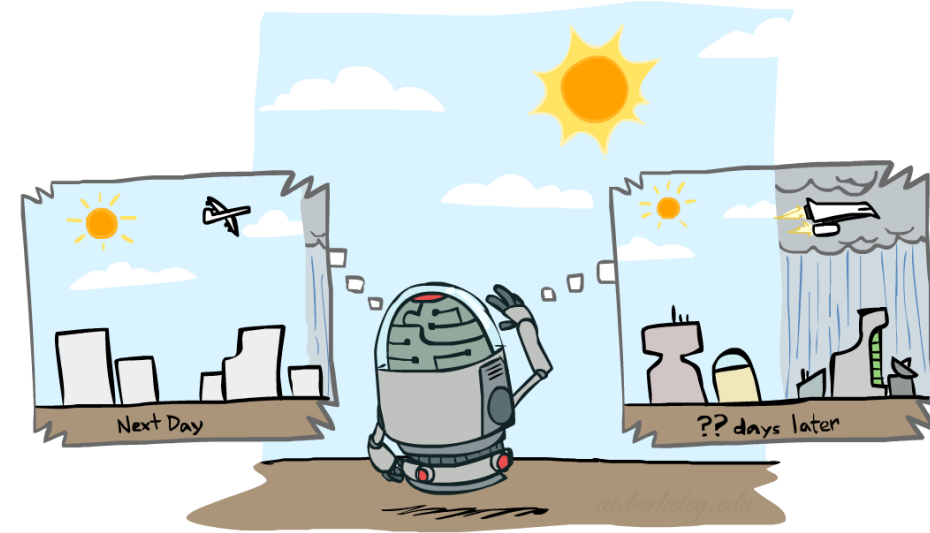
- Question: What's $P(X)$ on some day t ?



$$P(x_1) = \text{known}$$

$$\begin{aligned} P(x_t) &= \sum_{x_{t-1}} P(x_{t-1}, x_t) \\ &= \sum_{x_{t-1}} P(x_t \mid x_{t-1}) P(x_{t-1}) \end{aligned}$$

Forward simulation



Example Run of Mini-Forward Algorithm

- From initial observation of sun

$$\begin{array}{ccccc}
 \left\langle \begin{array}{c} 1.0 \\ 0.0 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.9 \\ 0.1 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.84 \\ 0.16 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.804 \\ 0.196 \end{array} \right\rangle & \longrightarrow \left\langle \begin{array}{c} 0.75 \\ 0.25 \end{array} \right\rangle \\
 P(X_1) & P(X_2) & P(X_3) & P(X_4) & P(X_\infty)
 \end{array}$$

- From initial observation of rain

$$\begin{array}{ccccc}
 \left\langle \begin{array}{c} 0.0 \\ 1.0 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.3 \\ 0.7 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.48 \\ 0.52 \end{array} \right\rangle & \left\langle \begin{array}{c} 0.588 \\ 0.412 \end{array} \right\rangle & \longrightarrow \left\langle \begin{array}{c} 0.75 \\ 0.25 \end{array} \right\rangle \\
 P(X_1) & P(X_2) & P(X_3) & P(X_4) & P(X_\infty)
 \end{array}$$

- From yet another initial distribution $P(X_1)$:

$$\begin{array}{ccc}
 \left\langle \begin{array}{c} p \\ 1 - p \end{array} \right\rangle & \dots & \longrightarrow \left\langle \begin{array}{c} 0.75 \\ 0.25 \end{array} \right\rangle \\
 P(X_1) & & P(X_\infty)
 \end{array}$$

Forward algorithm (simple form)

- What is the state at time t

- $P(X_t) = \sum_{x_{t-1}} P(X_t, X_{t-1}=x_{t-1})$
- $= \sum_{x_{t-1}} P(X_{t-1}=x_{t-1}) P(X_t | X_{t-1}=x_{t-1})$

Probability from
previous iteration

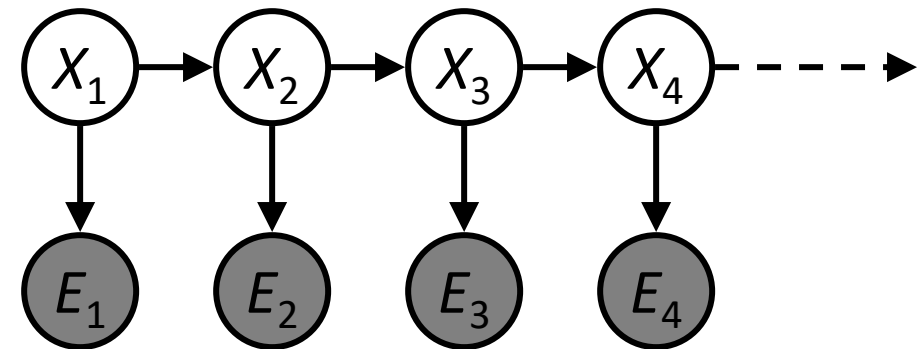
Transition model

- Iterate this update starting at $t=0$

- $P(X_1) = P(X_1)$
- $P(X_2) = P(X_1) P(X_2 | X_1)$
- $P(X_3) = P(X_2) P(X_3 | X_2)$
- $P(X_1, X_2, X_3) = P(X_1) P(X_2 | X_1) P(X_3 | X_2)$

Hidden Markov Models

- Markov chains not so useful for most agents
 - Need observations to update your beliefs
- Hidden Markov models (HMMs)
 - Underlying Markov chain over states X
 - You observe outputs (effects) at each time step
- An HMM is a temporal probabilistic model in which the state of the process is described by a single, discrete random variable
- HMMs require the state to be a single, discrete variable, there is no corresponding restriction on the evidence variables.



Example: Weather HMM

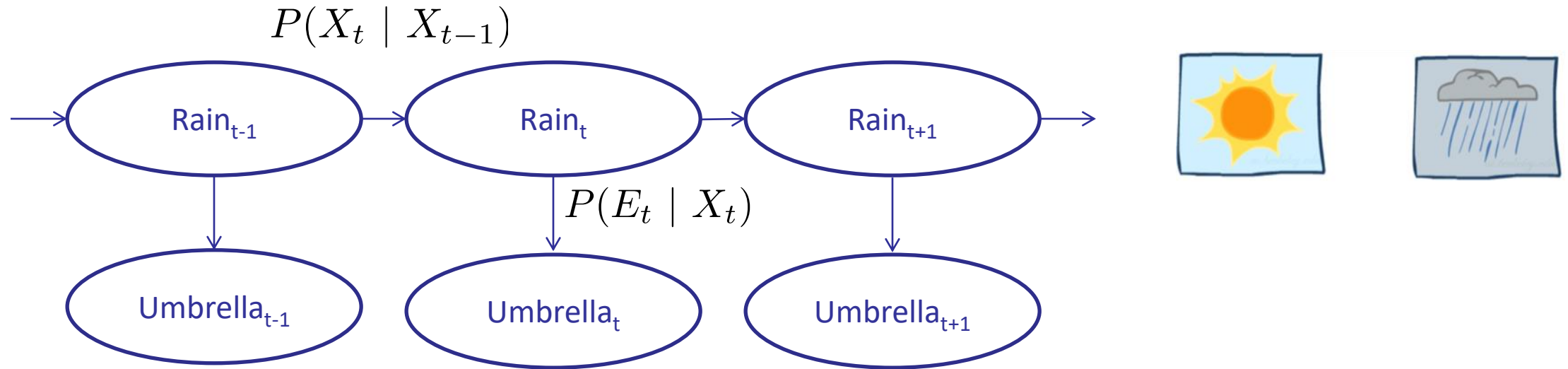


Figure 2: Bayesian network structure and conditional distributions describing the umbrella world. The transition model is $P(\text{Rain}_t \mid \text{Rain}_{t-1})$ and the sensor model is $P(\text{Umbrella}_t \mid \text{Rain}_t)$.

- An HMM is defined by: (Markov Chains + observed Variables)

- Initial distribution:
- Transitions:
- Emissions:

$$P(X_1)$$

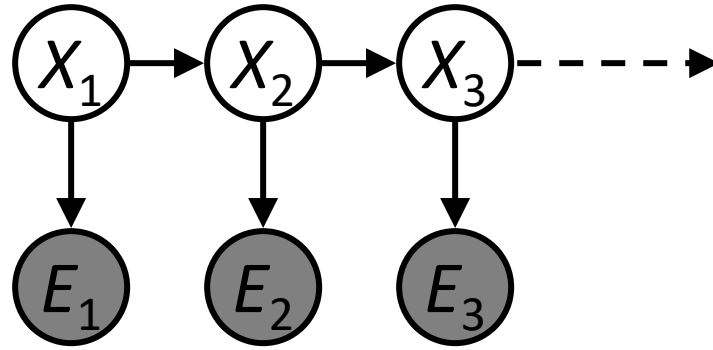
$$P(X_t \mid X_{t-1})$$

$$P(E_t \mid X_t)$$

R_{t-1}	R_t	$P(R_t \mid R_{t-1})$	R_t	U_t	$P(U_t \mid R_t)$
+r	+r	0.7	+r	+u	0.9
-r	+r	0.3	-r	+u	0.1

$$P(\mathbf{X}_{0:t}, \mathbf{E}_{1:t}) = P(\mathbf{X}_0) \prod_{i=1}^t P(\mathbf{X}_i \mid \mathbf{X}_{i-1}) P(\mathbf{E}_i \mid \mathbf{X}_i).$$

Formally Joint Distribution of an HMM



- Joint distribution

$$P(X_1, E_1, X_2, E_2, X_3, E_3) = P(X_1) P(E_1 | X_1) P(X_2 | X_1) P(E_2 | X_2) P(X_3 | X_2) P(E_3 | X_3)$$

- More generally

$$P(X_1, E_1, \dots, X_T, E_T) = P(X_1) P(E_1 | X_1) \prod_t P(X_t | X_{t-1}) P(E_t | X_t)$$

Example: Weather HMM



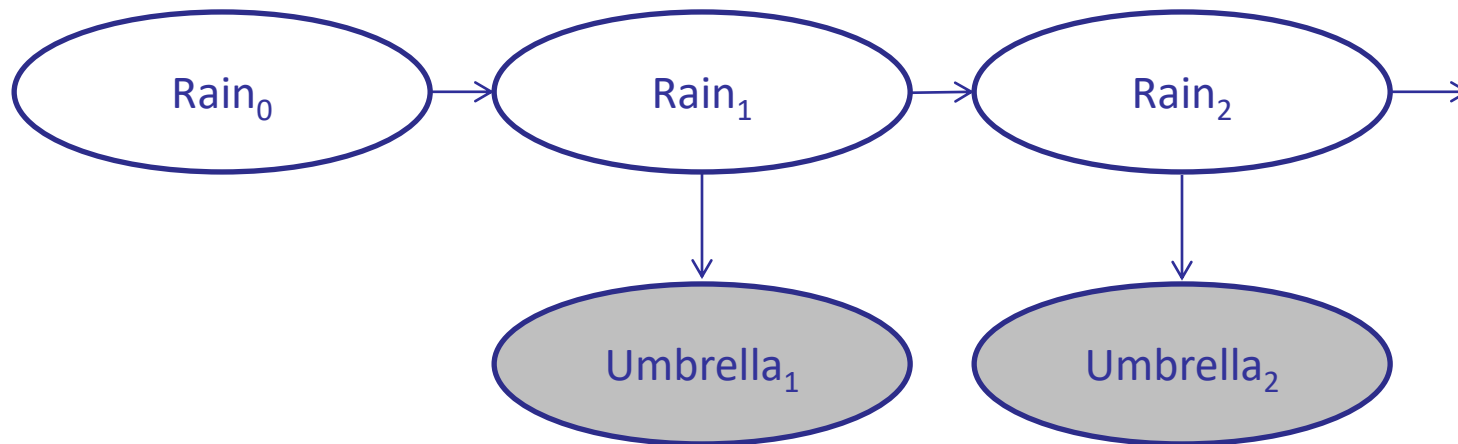
On day 0, we have no observations, only the security guard's prior beliefs; let's assume that consists of $P(R_0) = 0.5, 0.5$.

$$P(R_1) = P(R_0) P(R_1 | R_0)$$

$$B(+r) = 0.5$$

$$B(-r) = 0.5$$

$$P(R_1) = P(+R_0) P(+R_1 | +R_0) + P(-R_0) P(-R_1 | -R_0)$$



Transition Probabilities Emission Probabilities

R_t	R_{t+1}	$P(R_{t+1} R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

R_t	U_t	$P(U_t R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Example: Weather HMM

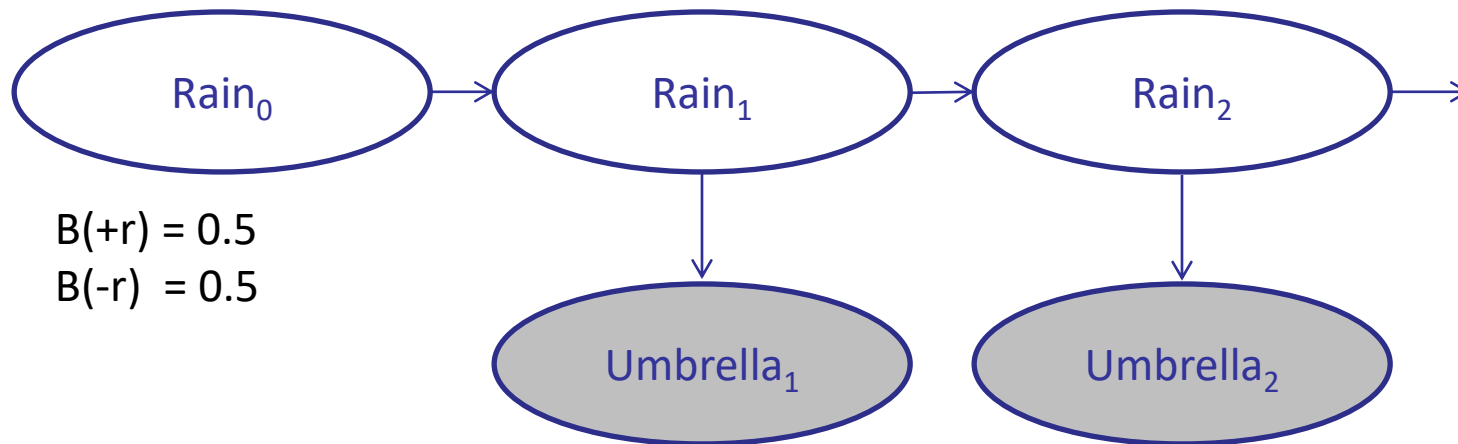


On day 1, the umbrella appears, so $U = \text{true}$, The prediction from $t = 0$ to $t = 1$ is

$$P(R_1) = \sum_{r_0} P(R_1 | r_0) P(r_0) = \langle 0.7, 0.3 \rangle \times 0.5 + \langle 0.3, 0.7 \rangle \times 0.5 = \langle 0.5, 0.5 \rangle$$

and updating it with the evidence for $t = 1$ gives

$$\begin{aligned} P(R_1 | u_1) &= a P(u_1 | R_1) P(R_1) = a \langle 0.9, 0.2 \rangle \langle 0.5, 0.5 \rangle \\ &= a \langle 0.45, 0.1 \rangle \approx \langle 0.818, 0.182 \rangle. \end{aligned}$$



Transition Probabilities Emission Probabilities

R_t	R_{t+1}	$P(R_{t+1} R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

R_t	U_t	$P(U_t R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Example: Weather HMM



On day 1, the umbrella appears, so $U = \text{true}$, The prediction from $t = 0$ to $t = 1$ is

$$P(R_1) = \sum_{r_0} P(R_1 | r_0) P(r_0) = \langle 0.7, 0.3 \rangle \times 0.5 + \langle 0.3, 0.7 \rangle \times 0.5 = \langle 0.5, 0.5 \rangle$$

and updating it with the evidence for $t = 1$ gives

$$\begin{aligned} P(R_1 | u_1) &= a P(u_1 | R_1) P(R_1) = a \langle 0.9, 0.2 \rangle \langle 0.5, 0.5 \rangle \\ &= a \langle 0.45, 0.1 \rangle \approx \langle 0.818, 0.182 \rangle . \end{aligned}$$

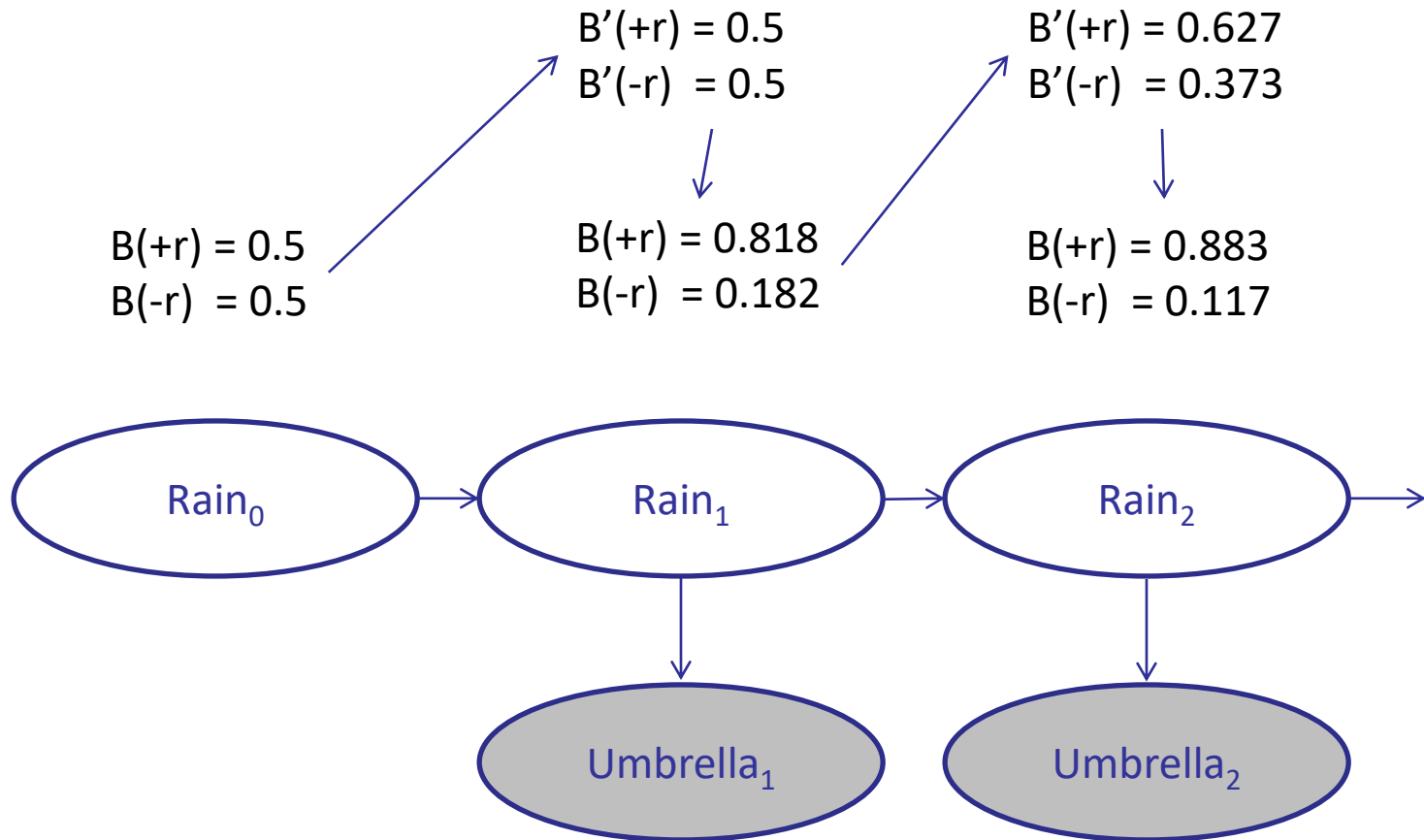
On day 2, the umbrella appears, so $U = \text{true}$, The prediction from $t = 1$ to $t = 2$ is

$$\begin{aligned} P(R_2 | u_1) &= \sum_{r_1} P(R_2 | r_1) P(r_1 | u_1) \\ &= \langle 0.7, 0.3 \rangle \times 0.818 + \langle 0.3, 0.7 \rangle \times 0.182 \approx \langle 0.627, 0.373 \rangle , \end{aligned}$$

and updating it with the evidence for $t = 2$ gives

$$\begin{aligned} P(R_2 | u_1, u_2) &= \alpha P(u_2 | R_2) P(R_2 | u_1) = a \langle 0.9, 0.2 \rangle \langle 0.627, 0.373 \rangle \\ &= a \langle 0.565, 0.075 \rangle \approx \langle 0.883, 0.117 \rangle . \end{aligned}$$

Example: Weather HMM

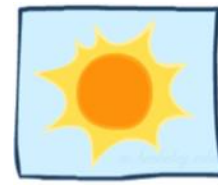


Transition Probabilities Emission Probabilities

R_t	R_{t+1}	$P(R_{t+1} R_t)$
$+r$	$+r$	0.7
$+r$	$-r$	0.3
$-r$	$+r$	0.3
$-r$	$-r$	0.7

R_t	U_t	$P(U_t R_t)$
$+r$	$+u$	0.9
$+r$	$-u$	0.1
$-r$	$+u$	0.2
$-r$	$-u$	0.8

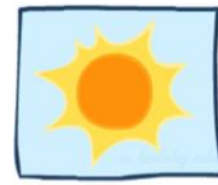
Example 2: Weather and Mode HMM



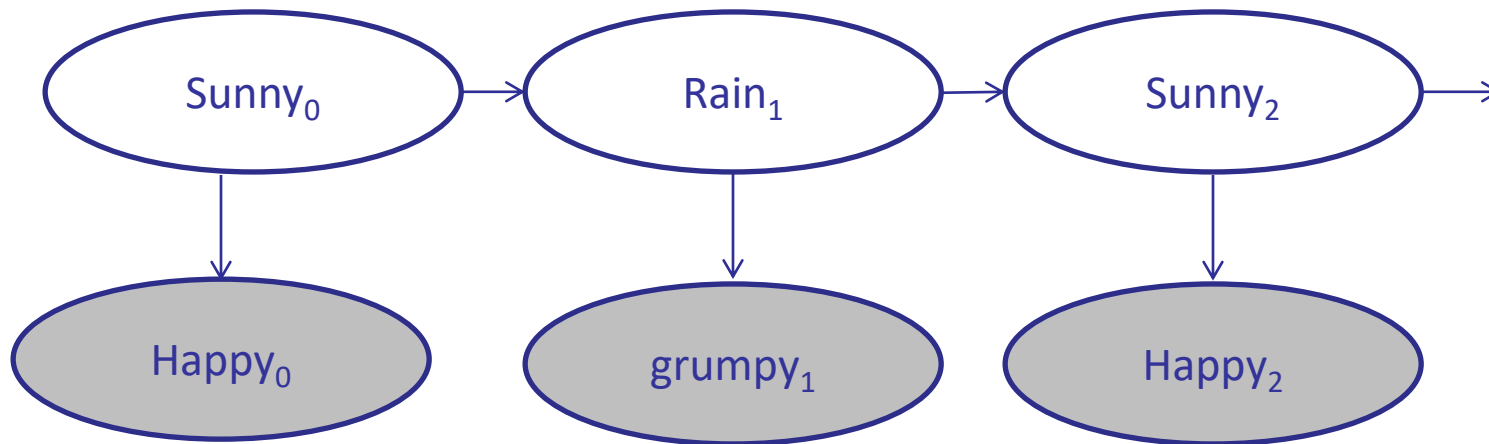
Example: Consider the example which elaborates how a person feels on different climates.



Example 2: Weather and Mode HMM



Example: Consider the example which elaborates how a person feels on different climates.



Example 2: Weather and Mode HMM



Example: Consider the example which elaborates how a person feels on different climates.



Transition Probabilities

	→		8	0.8
	→		2	0.2
	→		2	0.4
	→		3	0.6

Transition Probabilities

S_t	S_{t+1}	$P(S_{t+1} S_t)$
sunny	sunny	0.8
sunny	rainy	0.2
rainy	rainy	0.6
rainy	sunny	0.4

Example 2: Weather and Mode HMM



Example: Consider the example which elaborates how a person feels on different climates.



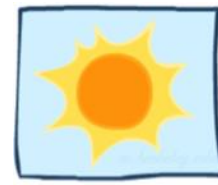
Emission Probabilities

	→		8	0.8
	→		2	0.2
	→		2	0.4
	→		3	0.6

Emission Probabilities

S_t	H_t	$P(H_t S_t)$
sunny	happy	0.8
sunny	grumpy	0.2
rainy	happy	0.4
rainy	grumpy	0.6

Example 2: Weather and Mode HMM



Example: Consider the example which elaborates how a person feels on different climates.



Probability of sunny



10 / 15

0.67

Probability of rainy



5 / 15

0.33

Probability of happy



10 / 15

0.67

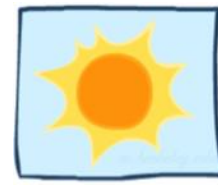
Probability of grumpy



5 / 15

0.33

Example 2: Weather and Mode HMM



If Happy today, what is probability its sunny or rainy?

$$P(\text{Sunny} | \text{Happy}) = P(\text{Happy} | \text{Sunny}) P(\text{sunny}) / P(\text{Happy}) \Rightarrow 0.8 * 0.67 / 0.67 \Rightarrow 0.8$$

$$P(\text{rainy} | \text{Happy}) = P(\text{Happy} | \text{rainy}) P(\text{rainy}) / P(\text{Happy}) \Rightarrow 0.4 * 0.33 / 0.67 = 0.2$$

S_t	H_t	$P(H_t S_t)$
sunny	happy	0.8
sunny	grumpy	0.2
rainy	happy	0.4
rainy	grumpy	0.6

S_t	S_{t+1}	$P(S_{t+1} S_t)$
sunny	sunny	0.8
sunny	rainy	0.2
rainy	rainy	0.6
rainy	sunny	0.4

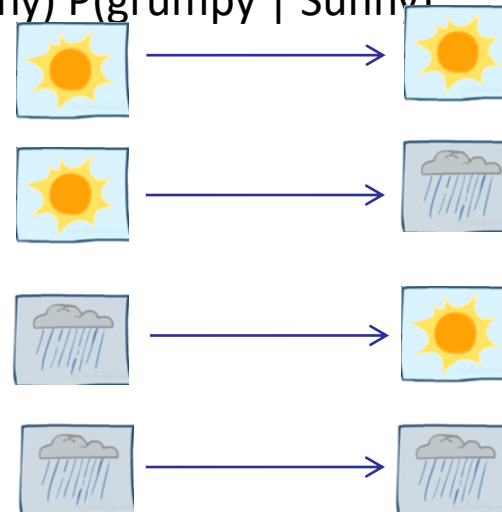
Example 2: Weather and Mode HMM



If Happy-grumpy, what is weather for 2 days?

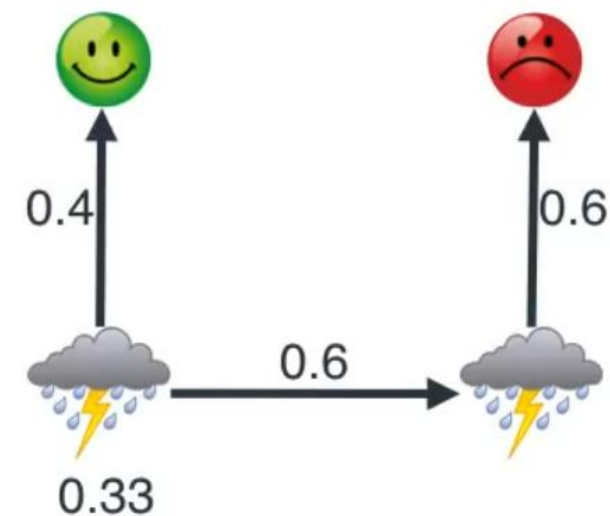
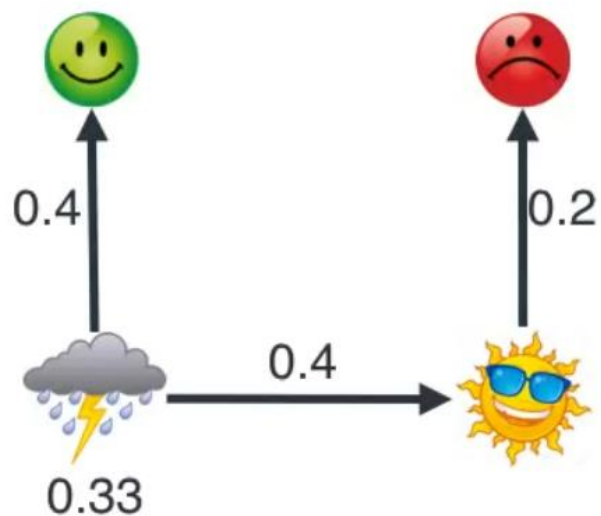
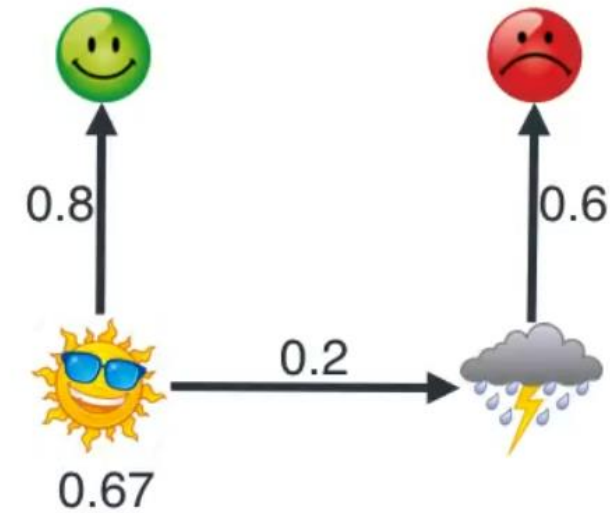
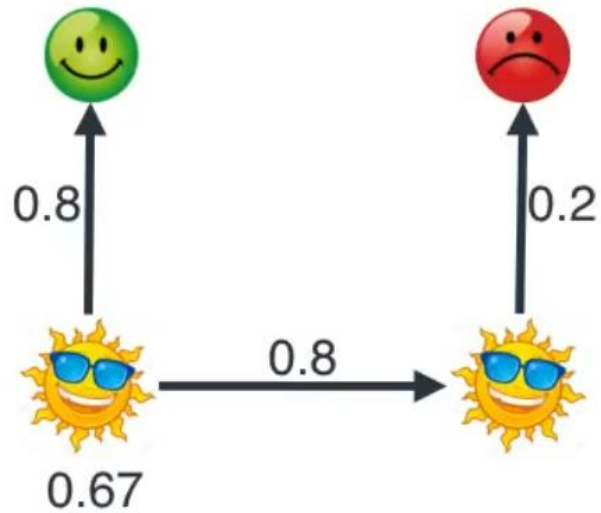
- $P(\text{Sunny, Rainy}) = P(\text{Sunny}) P(\text{Happy} \mid \text{Sunny}) P(\text{Rainy} \mid \text{Sunny}) P(\text{grumpy} \mid \text{Rainy})$
- $P(\text{Sunny, Rainy}) = 0.67 * 0.8 * 0.2 * 0.6 \Rightarrow 0.064$
- $P(\text{Sunny, Sunny}) = P(\text{Sunny}) P(\text{Happy} \mid \text{Sunny}) P(\text{Sunny} \mid \text{Sunny}) P(\text{grumpy} \mid \text{Sunny})$
- $P(\text{Sunny, Sunny}) = 0.67 * 0.8 * 0.8 * 0.2 \Rightarrow 0.085$
- $P(\text{Rainy, Sunny}) = P(\text{Rainy}) P(\text{Happy} \mid \text{Rainy}) P(\text{Rainy} \mid \text{Sunny}) P(\text{grumpy} \mid \text{Sunny})$
- $P(\text{Sunny, Rainy}) = 0.33 * 0.4 * 0.4 * 0.2 \Rightarrow 0.010$

S_t	H_t	$P(H_t \mid S_t)$
sunny	happy	0.8
sunny	grumpy	0.2
rainy	happy	0.4
rainy	grumpy	0.6



S_t	S_{t+1}	$P(S_{t+1} \mid S_t)$
sunny	sunny	0.8
sunny	rainy	0.2
rainy	rainy	0.6
rainy	sunny	0.4

Example 2: Weather and Mode HMM



Filtering / Monitoring

- Filtering, or monitoring, is the task of tracking the distribution $B_t(X) = P_t(X_t \mid e_1, \dots, e_t)$ (the belief state) over time
- We start with $B_1(X)$ in an initial setting, usually uniform
- As time passes, or we get observations, we update $B(X)$
- The Kalman filter was invented in the 60's and first implemented as a method of trajectory estimation for the Apollo program.
- With HMM infer discrete, finite variable and using Kalman filter we can have inference of continuous variables.